

# A Dense-Common-Domain Answer to the Weak Trotter–Kato Question

Literature-implied partial subcase of arXiv:0805.1027

Lane 19, agent\_lane\_19, model GPT5.6

**Status.** `literature_implied_answer (partial subcase)`. The unrestricted Banach-space question is not claimed solved.

**Source question.** Eisner and Serény, *On the weak analogue of the Trotter–Kato theorem*, arXiv:0805.1027, Question 2.6 on PDF page 3, consider  $C_0$ -semigroups  $T_n, T$  with a common exponential bound and generators  $A_n, A$ . For one (equivalently every)  $\lambda$  in the common right half-plane, their property (1) is

$$R(\lambda, A_n)^k x \rightharpoonup R(\lambda, A)^k x \quad (x \in X, k \in \mathbb{N}).$$

They ask whether this is equivalent to  $T_n(t)x \rightharpoonup T(t)x$  uniformly for  $t$  in compact intervals [1].

**Supporting theorem.** Lee, *A Convergence of  $C_0$  Semigroups in the Weak Topology*, Theorem 2 on PDF page 3, proves the following. Let  $T_n, T$  be contraction  $C_0$ -semigroups on an arbitrary Banach space. If the resolvents converge weakly to the resolvent of  $A$  and

$$D = \left\{ x \in \bigcap_{n \geq 1} D(A_n) : \sup_{n \geq 1} \|A_n x\| < \infty \right\}$$

is dense in  $X$ , then  $T_n(t)x \rightharpoonup T(t)x$  uniformly on bounded time intervals [2, Theorem 2].

**Identification.** The source paper observes immediately after property (1) that convergence of all powers at one  $\lambda$  implies weak resolvent convergence locally uniformly throughout the common right half-plane. It therefore satisfies Lee’s resolvent hypothesis. Lee’s theorem gives an affirmative answer to Question 2.6 for contraction semigroups having the displayed dense common-domain boundedness property. This implication is an identification of the two statements, not a new theorem; Lee explicitly cites Eisner–Serény and presents the result as weak Trotter–Kato convergence.

**Why the extra hypothesis matters in the proof.** For  $x \in D$ ,

$$\|T_n(h)x - x\| \leq \int_0^h \|T_n(s)A_n x\| ds \leq h \sup_n \|A_n x\|.$$

Thus scalar orbits are equicontinuous uniformly in  $n$  on a dense set. Weak convergence of their Laplace transforms then upgrades to uniform weak convergence on bounded time intervals, and density extends it to every  $x \in X$ . This pinpoints the missing ingredient in the unrestricted problem.

**Scope and search status.** The theorem does not remove the density condition and is stated for contractions. Bounded searches through 2026-08-13, using the exact question wording, title/authors, and the terms “weak Trotter–Kato” and “weak convergence of powers of resolvents”, found the Hilbert-space result of Marinoschi (2012), Lee’s uniformly-convex-dual extension (2013), and the general-Banach dense-domain theorem above, but no unrestricted proof or counterexample. Eight direct upgrade attempts are recorded in the accompanying run attempt; their common obstruction is uniform Yosida approximation without equicontinuity.

**Human-review recommendation.** Verify the quantifier on  $\lambda$  in Lee’s Theorem 2 and the source paper’s resolvent-series implication. Then accept this as a literature-implied partial subcase only, retaining the unrestricted question as open.

## References

- [1] T. Eisner and A. Serény, *On the weak analogue of the Trotter–Kato theorem*, Taiwanese J. Math. 14 (2010), 1411–1416; arXiv:0805.1027.
- [2] Y. S. Lee, *A Convergence of  $C_0$  Semigroups in the Weak Topology*, Korean J. Math. 22 (2014), 349–354, doi:10.11568/kjm.2014.22.2.349.